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Unconventional Floquet Quantum Spin Hall Effect in Two-Dimensional Altermagnets

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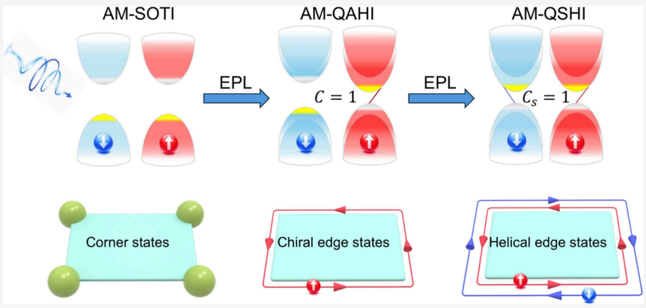

ABSTRACT: Altermagnetism has recently emerged as a promising platform for achieving exotic quantum states and dissipationless spin transport. Here, we uncover a previously unexplored Floquet realization of the quantum spin Hall effect in 2D altermagnets under elliptically polarized light, a phenomenon that has remained unattainable in previously studied ferromagnetic and antiferromagnetic systems. Remarkably, accompanying the periodic driving, the system undergoes successive topological phase transitions from a second-order topological insulator to a quantum anomalous Hall insulator, and ultimately to a quantum spin Hall insulator. Moreover, the proposed mechanism is confirmed in the altermagnetic V_2SSeO monolayer through analyses of corner states, edge states, and quantized anomalous and spin Hall responses, establishing it as a realistic material platform. Our findings establish a link between altermagnetism, Floquet engineering, and topological phases, opening new possibilities for next-generation optically controlled topological spintronic devices.

KEYWORDS: *altermagnetism, Floquet engineering, quantum spin Hall effect, topological phase transitions*

Altermagnetism, emerging as a novel fundamental class of magnetic order distinct from conventional ferromagnetism and antiferromagnetism, is in the focus of intensive research.[1−11] In altermagnets, oppositely oriented magnetic moments on different sublattices are connected by rotational or mirror symmetries, leading to fully compensated collinear magnetism in real space while simultaneously generating pronounced nonrelativistic spin splitting in momentum space.[12−22] Owing to this unique combination of ferromagnetic and antiferromagnetic characteristics, altermagnets provide a unique platform for realizing a variety of exotic phenomena, including spin-polarized transport, anomalous Hall and Nernst effects, magneto-optical responses, and chiral magnons.[23−31] More importantly, the integration of altermagnetism with topology has attracted widespread attentions, since the interplay between symmetry-driven spin splitting and nontrivial topological band structures not only enriches the family of unconventional topological quantum states, but also opens promising avenues toward dissipationless and highly efficient spintronic functionalities.[32−38]

Floquet engineering, on the other hand, provides a powerful paradigm for dynamically controlling electronic transport and band topology.[39,40] Experimentally, the periodic driving has been shown to induce anomalous Hall effects in graphene, while more recent time-resolved ARPES measurements have directly revealed Floquet−Bloch states and light-induced band gaps in monolayer graphene.[41−43] Moreover, with strong light−matter interaction, enormous Floquet topological states, such as the Floquet quantum anomalous Hall effect (QAHE), Floquet second-order topological insulator (SOTI), and Floquet Weyl semimetal, are proposed and intensively explored.[44−51] Despite the rapid progress in Floquet topological phases, the emergence of a Floquet quantum spin Hall insulator (QSHI) with a nontrivial Z_2 invariant and one pair of gapless edge states remains largely unexplored. Notably, as a paradigmatic topological state, the QSHI has played a pivotal role in deepening our understanding of topological quantum physics and material systems.[52−58] The realization of a Floquet QSHI would not only generalize the QSHI paradigm to nonequilibrium quantum systems, but also open new opportunities for ultrafast and noncontact optical control of topological phase transitions and spin transport through periodic driving.

In this work, we put forward that 2D altermagnets provides an exciting platform for realizing the Floquet QSHI under elliptically polarized light (EPL). In particular, the V_2SSeO monolayer, featuring intrinsic *d*-wave altermagnetic spin splitting together with a second-order topological insulator (SOTI) phase in equilibrium, is identified as a highly versatile candidate for engineering and tuning the predicted topological

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phases. As schematically illustrated in Figures 1(a) and 1(b), the system undergoes multiple topological phase transitions among a SOTI, a QAHI, and a QSHI. Under EPL irradiation, the degeneracy between the X and Y valleys for the V_2SSeO monolayer is lifted, resulting in successive valley-selective band inversions. The first inversion occurs at a single valley and induces a transition from a SOTI into a QAHI with $C = 1$ and one chiral edge mode. As the photon energy $\hbar\omega$ further increases, a second inversion emerges at the other valley within the opposite spin channel, contributing an opposite Chern number that compensates the first one and drives the total Chern number to $C = 0$. Importantly, this compensation does not destroy the nontrivial topology, but instead leads to the emergence of a QSHI phase with a quantized spin Hall response and helical edge states localized near the X and Y valleys. These results highlight EPL as a powerful tool for tailoring band topology in emergent altermagnets, opening new opportunities for ultrafast and optically controlled spintronic devices.

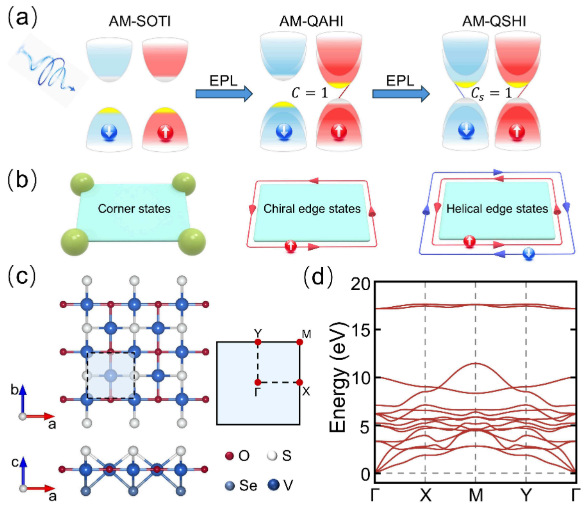

Figure 1. (a) Schematic of EPL-induced topological phase transitions in a 2D altermagnet, illustrating the progression from SOTI to QAHI and ultimately to a the QSHI. Red and blue bands indicate spin-up and spin-down channels, respectively. (b) Corresponding boundary states, including corner states, chiral edge states, and helical edge states. (c) Top and side views of the V_2SSeO monolayer, with the unit cell highlighted in light blue, and the corresponding two-dimensional Brillouin zone with labeled high-symmetry points. (d) Phonon spectrum of the V_2SSeO monolayer, indicating its dynamical stability.

The Janus V_2SSeO monolayer crystallizes in a square lattice with space group *P4mm* (No. 99) and point group symmetry C_{4v}. As displayed in Figure 1(c), the crystal is composed of three atomic layers along the *z* direction, with coplanar V and O atoms located between the outer S and Se layers. The optimized lattice constant is a = 3.88 Å. As depicted in Figure 1(d), the phonon spectrum shows no imaginary frequencies, confirming that the Janus V_2SSeO monolayer is dynamically stable. The valence configuration of V is $3d^34s^2$, but in the V_2SSeO monolayer, charge transfer to the surrounding O, S, and Se atoms yields an effective $3d^2$ configuration. Within the distorted octahedral crystal field, the V-3d orbitals split into t_{2g} and e_g states, with two parallel-spin electrons occupying separate t_{2g} orbitals and generating a magnetic moment of about 2 μ_B per V atom. To determine the magnetic ground state, we examine different magnetic configurations and find that the antiparallel alignment of magnetic moments between the two V atoms is energetically favored. Furthermore, upon including SOC, the magnetic anisotropy energy is calculated to be 0.05 meV per unit cell, with the easy axis along the in-plane diagonal direction.

In the absence of SOC, the V_2SSeO monolayer belongs to the spin space group $P\bar{1}41m\bar{1}m_{\infty m}1$ (No. 25.99. 1.1). The two V sublattices with opposite magnetic moments are related by the $\{C_2^{\perp}||C_4\}$ spin symmetry, enforcing zero net magnetization and establishing its altermagnetic nature. The pronounced nonrelativistic spin splitting emerges along Γ−X and Γ−Y, whereas bands along Γ−M remain spin-degenerate, exhibiting a characteristic *d*-wave form of spin splitting in momentum space, as illustrated in Figure 2(a) and Figure S1. Moreover,

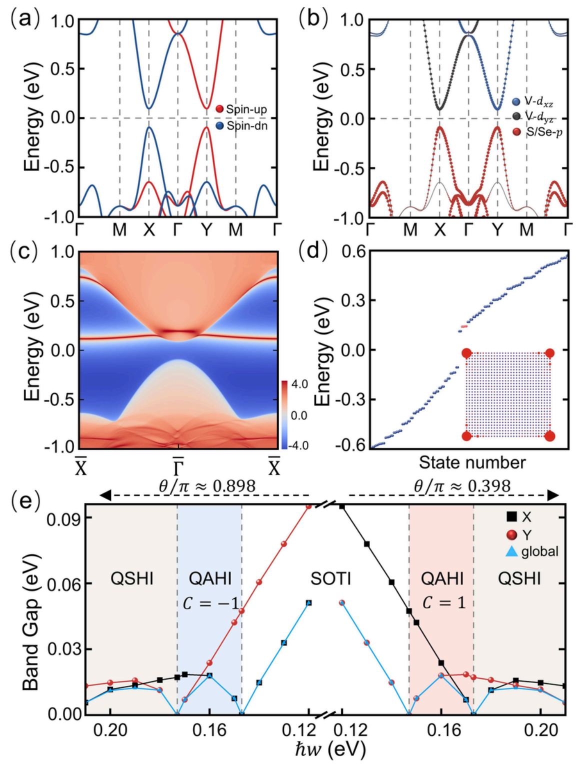

Figure 2. (a) Band structure of the altermagnetic V_2SSeO monolayer without SOC. Red and blue dots indicate spin-up and spin-down states, respectively. (b) Orbitally resolved band structure with SOC, weighted with the V-$d_{xz/yz}$ and S/Se-*p* orbitals. (c) Edge states of the altermagnetic V_2SSeO monolayer. (d) Energy spectrum of a tetragonal nanoflake of the V_2SSeO monolayer, with red markers denoting in-gap corner states. The inset shows the corresponding charge distribution of the corner states. (e) Phase diagram of the altermagnetic V_2SSeO monolayer under EPL irradiation at $\theta/\pi \approx 0.398$ and $\theta/\pi \approx 0.898$, as a function of photon energy $\hbar\omega$.

the V_2SSeO monolayer is a direct-gap insulator with a gap of 0.189 eV at the X and Y points, and the bands around the Fermi level are predominantly composed of V-$d_{xz/yz}$ and S/Se-*p* orbitals. Upon inclusion of SOC, with the magnetization aligned along the [110] direction, the symmetry of V_2SSeO is reduced from the spin space group to the magnetic space group $Cm'm2'$ (No. 35.167), which preserves the mirror symmetry M_{xy}. As presented in Figure 2(b), the overall band dispersion remains nearly unchanged, retaining a direct band gap of 0.185 eV without the SOC-induced band inversion. Therefore, the V_2SSeO monolayer cannot be a conventional

2D magnetic TI. However, a distinctive floating edge state is observed in Figure 2(c), serving as a hallmark of a SOTI. To further reveal the corner states of V_2SSeO, the energy spectrum of a tetragonal nanoflake, obtained from maximally localized Wannier functions (MLWFs) based on O-p, S-p, Se-p, and V-d orbitals, is presented in Figure 2(d). Clearly, four continuous energy states (red dots) are observed, with their corresponding charge densities localized at the four corners of the nanoflake, confirming the SOTI phase in the altermagnetic V_2SSeO monolayer.

In the V_2SSeO monolayer, the X and Y valleys are connected by the M_{xy} mirror symmetry and retain energetic degeneracy despite possessing opposite spin polarizations. Notably, the in-plane anisotropy of EPL breaks the M_{xy} symmetry, lifts the valley degeneracy and enables successive valley-selective band inversions, thereby providing an ideal platform for valley-selective topological control. The periodic optical driving is described by the vector potential $\mathbf{A}(t) = (A_x \cos(\omega t), \eta A_y \sin(\omega t), 0)$. Here, ω is the driving frequency, $\hbar\omega$ is the photon energy, and $\eta = \pm 1$ denote the handedness of the EPL. A_x and A_y are the field amplitudes along the x and y directions, respectively. The EPL is further characterized by the amplitude $k_A = eA_0/\hbar = e\sqrt{A_x^2 + A_y^2}/\hbar$ and the polarization angle $\theta = \text{atan2}(A_y, A_x)$. Within the Peierls substitution, the hopping amplitudes acquire a time-dependent phase factor $t_{ij}(t) = t_{ij} \exp\left[i\frac{e}{\hbar}\mathbf{A}(t)\cdot\mathbf{d}_{ij}\right]$, where $\mathbf{d}_{ij}$ denotes the hopping vector between sites i and j. The periodically driven Hamiltonian $H(\mathbf{k}, t) = H(\mathbf{k}, t + T)$ can be solved within the Floquet formalism.[59−61] By transforming into the frequency domain, one obtains the time-independent Floquet Hamiltonian

$$H^F_{nm}(\mathbf{k}) = n\hbar\omega\delta_{nm} + h_{n-m}(\mathbf{k}), \tag{1}$$

where the Fourier components are defined as

$$h^F_{mn}(\mathbf{k}) = \frac{1}{T}\int_0^T dt H(\mathbf{k}, t) e^{i\omega(m-n)t}, \tag{2}$$

with $T = 2\pi/\omega$ being the driving period. The Floquet−Bloch states $|u^n_{\mathbf{k}}\rangle$ and the quasienergies $\varepsilon_n(\mathbf{k})$ are obtained by diagonalizing the Floquet Hamiltonian

$$H^F(\mathbf{k})|u^n_{\mathbf{k}}\rangle = \varepsilon_n(\mathbf{k})|u^n_{\mathbf{k}}\rangle. \tag{3}$$

Taking the $\theta/\pi \approx 0.398$ and $k_A \approx 0.095$ Å^{-1} as an example, Figure 2(e) presents the band gaps of the V_2SSeO monolayer as a function of the photon energy $\hbar\omega$, together with the corresponding evolution of the spin-resolved band structure near the X and Y points in Figure S2(a). As expected, both the band gap and the associated nontrivial band topology can be effectively engineered by the photon energy $\hbar\omega$. Specifically, owing to the EPL-induced lifting of the valley degeneracy, the gaps at the X and Y points undergo sequential, rather than simultaneous, closing and reopening, giving rise to a rich topological phase diagram. At photon energy $\hbar\omega = 0.147$ eV, a critical state emerges, in which the spin-up bands close at the Y point while the spin-down bands at the X point remain gapped. Upon further increasing photon energy $\hbar\omega$, the gap at Y reopens, signaling a topological phase transition. As illustrated in Figure 3(a), orbital-resolved analysis indicate that, compared with the SOTI phase, the highest occupied band and the lowest unoccupied band at the Y point are predominantly derived from V-d_{xz} and S/Se-p orbitals, respectively. Consequently, a band inversion emerges at the Y point, while the X point remains uninverted.

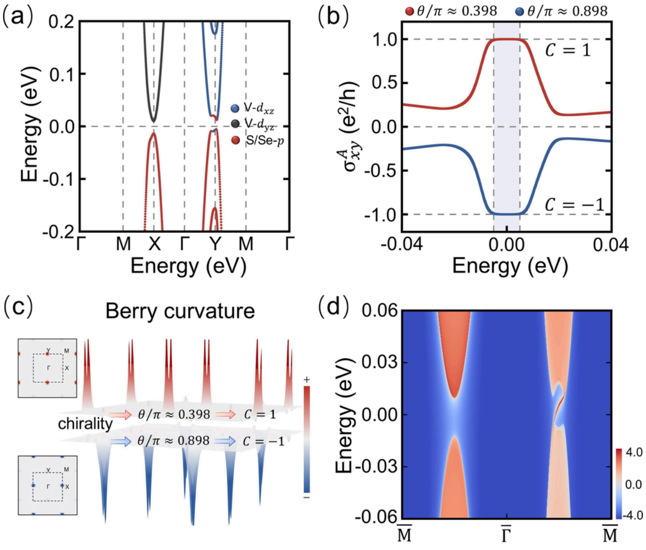

Figure 3. (a) Orbital-resolved band structure of the V_2SSeO monolayer under EPL irradiation with $\theta/\pi \approx 0.398$ at a photon energy of $\hbar\omega = 0.16$ eV. (b) Energy dependence of the anomalous Hall conductivity σ^A_{xy} for the V_2SSeO monolayer under EPL irradiation with $\theta/\pi \approx 0.398$ and $\theta/\pi \approx 0.898$. The anomalous Hall conductances within the nontrivial band gaps are quantized, corresponding to $C = \pm 1$, respectively. (c) k-space distribution of the Berry curvature for the V_2SSeO monolayer under EPL irradiation with $\theta/\pi \approx 0.398$ and $\theta/\pi \approx 0.898$. (d) Topological chiral edge states for the V_2SSeO monolayer under EPL irradiation with $\theta/\pi \approx 0.398$ at $\hbar\omega = 0.16$ eV.

To establish the nontrivial topology of the obtained insulating phase, the anomalous Hall conductivity is calculated using $\sigma^A_{xy} = Ce^2/h$, where C is the Chern number, defined as $C = \frac{1}{2\pi}\int_{BZ} \Omega(\mathbf{k}) d^2k$, and the Berry curvature over the occupied states is given by[62,63]

$$\Omega(\mathbf{k}) = -\sum_{n<E_F}\sum_{m\neq n} 2\text{Im}\frac{\langle\psi_{n\mathbf{k}}|v_x|\psi_{m\mathbf{k}}\rangle\langle\psi_{m\mathbf{k}}|v_y|\psi_{n\mathbf{k}}\rangle}{(\varepsilon_{m\mathbf{k}} - \varepsilon_{n\mathbf{k}})^2}, \tag{4}$$

where $\psi_{m/n\mathbf{k}}$ are the Bloch wave functions of bands m/n, with $\varepsilon_{m/n\mathbf{k}}$ being the corresponding eigenvalues and $v_{x/y}$ denoting the velocity operators. Figures 3(b) and 3(c) present the energy-dependent anomalous Hall conductivity σ^A_{xy} and the k-space distribution of the Berry curvature $\Omega(\mathbf{k})$ for the V_2SSeO monolayer with $\hbar\omega = 0.16$ eV. Within the band gap, σ^A_{xy} displays a quantized plateau, accompanied by Berry curvature $\Omega(\mathbf{k})$ mainly localized around the Y point, resulting in a Chern number of $C = 1$ and establishing the emergence of the Floquet QAHE. In addition, as illustrated in Figure S3 and Figure 3(d), the evolution of the Wannier charge centers (WCCs) together with the presence of one chiral edge state within the band gap further explicitly confirms the nontrivial QAHE phase in the altermagnetic V_2SSeO monolayer.

Remarkably, as the photon energy $\hbar\omega$ increases further, a second band inversion is triggered at the X point, accompanied by an inversion of orbital character, as shown in Figures 2(e) and 4(a). As a consequence of the band inversion taking place

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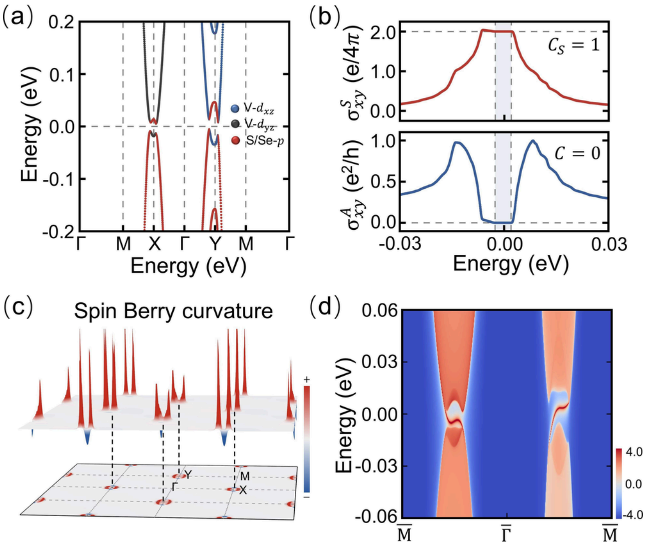

Figure 4. (a) Orbital-resolved band structure of the V_2SSeO monolayer. (b) Energy dependence of the spin Hall conductivity σ^S_{xy} and anomalous Hall conductivity σ^A_{xy}, showing a quantized spin Chern number of $C_s = 1$, while the total Chern number remains $C = 0$ within the nontrivial band gap. (c) Spin Berry curvature distribution of the occupied bands in the 2D Brillouin zone. (d) Edge states of the V_2SSeO monolayer, exhibiting the characteristic signatures of the QSHI phase. All results are obtained under EPL irradiation with $\theta/\pi \approx 0.398$ at a photon energy of $\hbar\omega$ = 0.19 eV.

in the opposite spin channel, the Berry curvature $\Omega(\mathbf{k})$ at the X point changes sign and becomes opposite to that at Y, as depicted in Figure S4. Hence, the two spin channels of the V_2SSeO monolayer carry opposite nonzero Chern numbers, i.e., $C_+ = +1$ and/or $C_- = -1$, which cancel each other in the total Chern number, $C = C_+ + C_- = 0$, while yielding a finite spin Chern number $C_S = (C_+ - C_-)/2 = 1$. The nonzero C_S is further confirmed by the calculations of spin Hall conductivity σ^S_{xy}, given by $\sigma^S_{xy} = e\hbar\int\frac{d^2k}{(2\pi)^2}\Omega^S(\mathbf{k})$.[64,65] The integrand $\Omega^S(\mathbf{k})$ is the so-called spin Berry curvature

$$\Omega^S(\mathbf{k}) = -2\mathrm{Im}\sum_{m\neq n}\frac{\langle\psi_{m\mathbf{k}}|J^S_x|\psi_{n\mathbf{k}}\rangle\langle\psi_{n\mathbf{k}}|v_y|\psi_{m\mathbf{k}}\rangle}{(E_{n\mathbf{k}} - E_{m\mathbf{k}})^2}. \quad (5)$$

Here, the spin current operator is defined as $J^S_x = \frac{\hbar}{4}\{\sigma^z, v_x\}$, describing a spin current flowing along the x direction with spin polarization perpendicular to the plane. As displayed in Figures 4(b) and 4(c), with $\hbar\omega$ = 0.19 eV as an example, σ^S_{xy} exhibits a quantized plateau in the insulating regime, originating predominantly from the pronounced spin Berry-curvature contributions around the X and Y points. This quantized conductance is in excellent agreement with $C_S = 1$ through the relation $\sigma^S_{xy} = C_S e/(2\pi)$, thereby demonstrating the realization of the QSHE. Moreover, one defining feature of QSHE is the existence of one pair of gapless edge states in the insulating gap. As illustrated in Figure 4(d), the edge states of a semi-infinite nanoribbon exhibit counterpropagating modes connecting the valence and conduction bands, providing unambiguous evidence for EPL-induced QSHE in the altermagnetic V_2SSeO monolayer.

Furthermore, a unique advantage of EPL lies in its ability to selectively tune band inversions in momentum space via the polarization angle. Combined with the spin-dependent band structure of altermagnets, this enables reversible switching of QAHI chirality while allowing the QSHE to remain robustly accessible. To verify this, we tune the polarization angle to $\theta/\pi \approx 0.898$, where, although the overall band structure remains nearly unchanged, the band gap first closes and then reopens at the X point in the opposite spin channel relative to $\theta/\pi \approx 0.398$, as depicted in Figure 2(e) and Figure S2(b). A Chern number of $C = -1$ is indeed obtained, as shown in Figure 3(b,c) and Figures S3 and S5. Remarkably, a QSHI phase continues to appear at higher photon energies, resulting from a second band inversion at the Y point in the opposite spin channel, as illustrated in Figures S2(b) and S6.

In summary, we have theoretically demonstrated that EPL provides a feasible dynamic route to realizing the QSHE in 2D altermagnets. Based on first-principles calculations, the proposed mechanism is further confirmed in the d-wave altermagnetic V_2SSeO monolayer, which naturally hosts a SOTI phase and opposite spin polarizations at the X and Y valleys. Remarkably, by breaking the M_{xy} mirror symmetry through its intrinsic in-plane anisotropy, EPL lifts the degeneracy between the X and Y valleys and, with increasing photon energy $\hbar\omega$, induces successive band inversions in opposite spin channels, thereby driving multiple topological phase transitions from a SOTI to a QAHI and ultimately to a QSHI. The nontrivial topological character of the altermagnetic V_2SSeO monolayer across these distinct phases is confirmed through analyses of corner states, edge states, as well as quantized anomalous and spin Hall conductivities. Our work opens a new Floquet avenue for realizing the QSHE in 2D altermagnets, while highlighting EPL as an effective means for controlling spin-resolved band topology and topological phase transitions.

METHODS

As implemented in the Vienna ab initio simulation package (VASP), the projector-augmented wave method was employed to perform the first-principles calculations based on density functional theory.[66,67] The generalized gradient approximation (GGA) of Perdew−Burke−Ernzerhof (PBE) was used for the exchange−correlation potential.[68] The plane-wave cutoff energy was set to 500 eV, and the Brillouin zone was sampled using a 11 × 11 × 1 Γ-centered k-point mesh. A vacuum layer of 20 Å was used to avoid interactions between the nearest slabs. The atomic positions and lattice constants were fully relaxed until the Hellmann−Feynman forces on each atom were smaller than 0.01 eV/Å. The electronic convergence criteria was set to 10^{-6} eV. To describe the localized 3d orbitals of the V atom, the GGA +U method was adopted with an effective Hubbard parameter U_{eff} = 3 eV. The maximally localized Wannier functions (MLWFs) are constructed using the WANNIER90 code.[69]

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.nanolett.6c02856.

Computational details of the Floquet Hamiltonian; orbital- and spin-resolved band structures; additional topological analyses; robustness tests; and Wannier interpolation (PDF)

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Notes

The authors declare no competing financial interest.

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